

Program : Diploma in Information Technology	
Course Code : 3262	Course Title: Fundamentals of System Administration
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3(L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To provide the basic concept of the operating system and its functions.
- To develop programs using shell scripting.
- To use Linux utilities and shell scripting to solve problems.

Course Pre-requisites:

Topic	Course Code	Course Name	Semester
Basic Knowledge of IT systems		Introduction to IT system	1
Control structure and loop		Problem Solving & Programming	2

Course Outcomes:

On completion of the course, students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Outline the basic concept of system software, operating system and program execution.	10	Understanding
CO2	Illustrate Process Concepts, Ideas in Memory Management, I/O Systems, file and Directory Management.	12	Understanding
CO3	Illustrate the Concept of Linux Commands and Basics of Shell Scripting.	12	Understanding
CO4	Summarize the Concept of Shell Scripting in Linux to solve problems.	9	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Outline the Basic Concept of System Software, Operating System And Program Execution.		
M1.01	Recall the concepts of Software, Application software operating system and System software.	1	Remembering
M1.02	Explain Usage of operating system.	1	Understanding
M1.03	Classify the operating system services.	1	Understanding
M1.04	Classify different types of operating systems.	2	Understanding
M1.05	Explain program execution.	1	Understanding
M1.06	List the I/O operation.	1	Remembering
M1.07	Illustrate how the file system manipulation are done.	1	Understanding
M1.08	Recall communication.	1	Remembering
M1.09	Explain error detection.	1	Understanding
Contents: System Software: Basic software system classification-Application software, System Software. Operating System: Operating System Overview, Usage and Services of Operating System, Different type of Operating Systems. Program Execution: Program Execution Basics, I/O Operations, File System Manipulation, Communication, Error detection.			
CO2	Illustrate Process Concepts, Ideas in Memory Management, I/O Systems, file and Directory Management.		
M2.01	Recall process	1	Remembering

M2.02	Explain process state and Process Control Block	1	Understanding
M2.03	Explain the need for memory management	1	Understanding
M2.04	Explain the concept and need of paging	1	Understanding
M2.05	Summarize segmentation	1	Understanding
M2.06	Explain virtual memory concept	1	Understanding
M2.07	Explain the Basic of input output system	1	Understanding
M2.08	Summarize file concept	1	Understanding
M2.09	Compare different file access methods	2	Understanding
M2.10	Classify various directory structures	2	Understanding
	Series Test – I	1	

Contents:

Process Management: Process Concept, Process States, Process Control Block

Memory Management: Need for Memory Management, Concept and need of Paging, segmentation, virtual memory. **I/O System:** Basics **File System:** file concept, different file access methods- (Direct access method, Sequential access method, Index access method), Directory Structures- (Single level directory structure, Two level directory structure, Tree directory structure)

CO3	Illustrate the Concept of Linux Commands and Basics of Shell Scripting.		
M3.01	Recall Directory management commands and its needs	2	Remembering
M3.02	Explain the different file management commands.	2	Understanding
M3.03	Explain the General-purpose command and need for them.	2	Understanding
M3.04	Explain shell.	1	Understanding
M3.05	Summarize kernel.	1	Understanding
M3.06	Outline shell prompt.	1	Understanding
M3.07	Explain shell scripting and advantages of shell script.	2	Understanding
M3.08	Summarize Linux environment variable	1	Understanding

Contents:

Directory Management Commands: ls, cd, pwd, mkdir, rmdir

File Management Commands: cat, chmod, cp, mv, rm,

General Purpose Commands: wc, od, lp, cal, who

Shell Scripting: shell, kernel, shell prompt, shell scripting, Advantage of shell script, Linux environment variable

CO4	Summarize the Concept of Shell Scripting in Linux to solve problems.		
M4.01	Compare basic operators and its needs in shell scripting.	2	Understanding
M4.02	Recall the concept of Decision making and looping structures in shell scripting	2	Remembering

M4.03	Illustrate shell looping concept	2	Understanding
M4.04	Summarize loop controls in shell scripting	3	Understanding
	Series Test – II	1	
Contents: Basic Operators: Arithmetic Operators, Relational Operators and Boolean Operators. Decision Making: if else statement Shell loop: while loop and for loop Loop Control: break and continue statement.			

Text / Reference

T/R	Book Title/Author
T1	Silverschatz, Galvin, Gagne “Operating System Concepts” 6th edition
R1	Richard Blum and Christine Bresnahan “Linux command line and shell scripting Bible”

Online Resources:

SLNo	Website Link
1	http://www.dbit.ac.in/cse/syllabus/system-administration-lab.pdf
2	https://www.linux.com/what-is-linux/
3	https://www.redhat.com/en/topics/linux/what-is-linux